

Exercise 1: SQL FUNDAMENTALS

① SELECT *
FROM employees;

ID	First-name	Last-name	department	Salary	hire-date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Molokane	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Maidoo	IT	72 000	2017-09-30	Johannesburg

② SELECT DISTINCT department
FROM employees;

Department
IT
Finance
HR
Marketing

③ SELECT first-name,
Last-name, salary
FROM employee
ORDER BY salary DESC;

First-name	Last-name	Salary
Eva	Maidoo	72 000
Alice	Green	70 000
Cathy	Zulu	65 000

David	Mokoena	50 000
Brian	Lee	45 000

④ SELECT id,
first_name,
last_name,
salary

FROM employees

WHERE salary > 60 000

ORDER BY salary DESC

LIMIT 3;

id	First_name	Last_name	Salary
5	Eva	Maido	72 000
1	Alice	Green	70 000
3	Carthy	Zulu	65 000

⑤

⑤ SELECT id,
first_name,
last_name, department

FROM employees

WHERE department = 'IT';

ID	First_name	Last_name	Department
1	Alice	Green	IT
5	Eva	Maido	IT

⑥ SELECT id,
first_name,
last_name,
department,
salary

FROM employees

WHERE department = 'Finance' AND salary > 60

id	First_name	Last_name	department	Salary
3	Carthy	Zulu	Finance	65 000

⑦ SELECT id,
First_name,
Last_name,
department

FROM employees

WHERE department = 'HR' or 'Marketing';

ID	First_name	Last_name	Department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

⑧ SELECT id,
First_name,
Last_name,
department

FROM employees

WHERE department NOT IN (IT);

ID	First_name	Last_name	department
2	Brian	Lee	HR
3	Carthy	Zulu	Finance
4	David	Mokoena	Marketing

⑨ SELECT id,
First_name,
Last_name,
department

FROM employees

WHERE department IN (HR, IT, finance);

id	First_name	Last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Maaido	IT

⑩ SELECT id,
First_name,
Last_name,
department,
Salary,
City

FROM employees

WHERE department = 'IT' AND salary > 65000 AND
city IN ('Johannesburg');

ID	first_name	Last_name	department	Salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Maaido	IT	72000	Johannesburg